

Phase I/II study of COVID-19 RNA vaccine BNT162b1 in adults

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In March 2020, the World Health Organization (WHO) declared coronavirus disease 2019 (COVID-19), which is caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2)¹, a pandemic. With rapidly accumulating numbers of cases and deaths reported globally², a vaccine is urgently needed. Here we report the available safety, tolerability and immunogenicity data from an ongoing placebo-controlled, observer-blinded dose-escalation study (ClinicalTrials.gov identifier NCT04368728) among 45 healthy adults (18–55 years of age), who were randomized to receive 2 doses—separated by 21 days—of 10 µg, 30 µg or 100 µg of BNT162b1. BNT162b1 is a lipid-nanoparticle-formulated, nucleoside-modified mRNA vaccine that encodes the trimerized receptor-binding domain (RBD) of the spike glycoprotein of SARS-CoV-2. Local reactions and systemic events were dose-dependent, generally mild to moderate, and transient. A second vaccination with 100 µg was not administered because of the increased reactogenicity and a lack of meaningfully increased immunogenicity after a single dose compared with the 30-µg dose. RBD-binding IgG concentrations and SARS-CoV-2 neutralizing titres in sera increased with dose level and after a second dose. Geometric mean neutralizing titres reached 1.9–4.6-fold that of a panel of COVID-19 convalescent human sera, which were obtained at least 14 days after a positive SARS-CoV-2 PCR. These results support further evaluation of this mRNA vaccine candidate.

In December 2019, a pneumonia outbreak of unknown cause occurred in Wuhan, China. By January 2020, a new coronavirus was identified as the aetiological agent. Within a month, the genetic sequence of the virus became available (MN908947.3). Infections with SARS-CoV-2 and the resulting disease, COVID-19, have spread globally. On 11 March 2020, the WHO declared the COVID-19 outbreak a pandemic¹. So far, the United States has reported the highest number of cases globally^{2,3}. No vaccines are currently available to prevent SARS-CoV-2 infection or COVID-19.

The RNA vaccine platform has enabled rapid vaccine development in response to this pandemic. RNA vaccines provide flexibility in the design and expression of vaccine antigens that can mimic the structure and expression of the antigen during natural infection. RNA is required for protein synthesis, does not integrate into the genome, is transiently expressed, is metabolized and eliminated by the natural mechanisms of the body and is therefore considered safe^{4–7}. RNA-based prophylactic infectious-disease vaccines and RNA therapeutic agents have been shown to be safe and well-tolerated in clinical trials. In general, vaccination with RNA elicits a robust innate immune response. RNA directs the

expression of the vaccine antigen in host cells and has intrinsic adjuvant effects⁸. A strength of the RNA-vaccine manufacturing platform—irrespective of the encoded pathogen antigen—is the ability to rapidly produce large quantities of vaccine doses against a new pathogen^{9,10}.

Vaccine RNA can be modified by incorporating 1-methylpseudouridine, which dampens innate immune sensing and increases mRNA translation *in vivo*¹¹. The BNT162b1 vaccine candidate that is currently investigated clinically incorporates such nucleoside-modified mRNA and encodes the RBD of the spike protein of SARS-CoV-2, a key target of virus-neutralizing antibodies^{12–14}. The RBD antigen expressed by BNT162b1 is modified by the addition of a T4 fibrinogen-derived foldon trimerization domain to increase its immunogenicity¹⁵ by multivalent display¹⁶. The proper folding of the RBDs in the resulting protein construct has been confirmed by high resolution structural analysis (A.B.V. et al., manuscript in preparation). The vaccine RNA is formulated in lipid nanoparticles for more-efficient delivery into cells after intramuscular injection¹⁷. BNT162b1 is one of several RNA-based SARS-CoV-2 vaccine candidates¹⁸ that are studied in parallel for selection to advance

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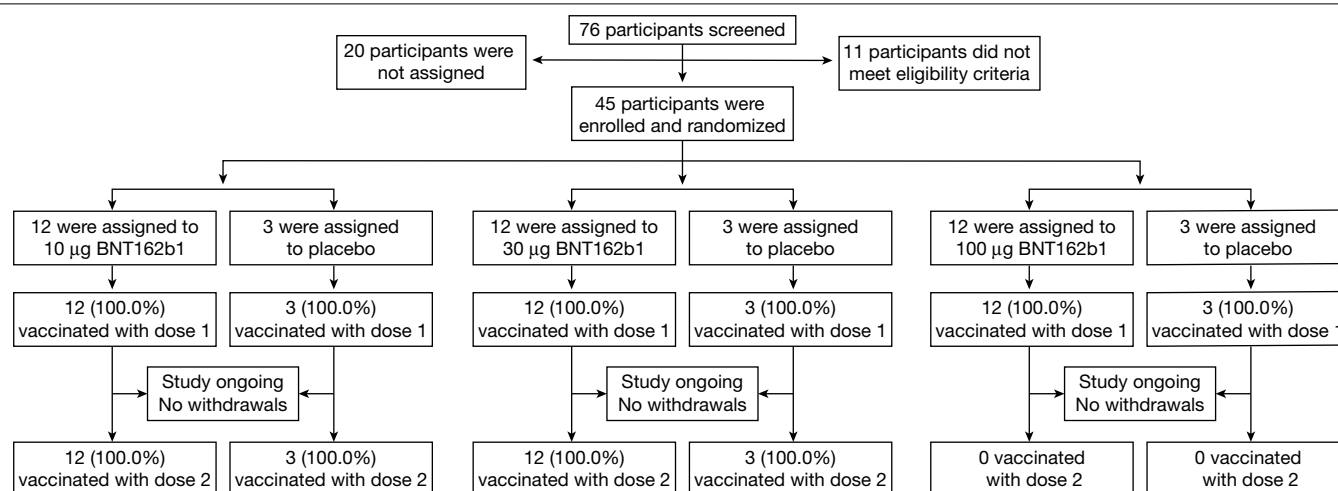


Fig. 1 | Study design. Participants who were not assigned ($n = 20$) were screened but not randomized because enrolment had closed.

to a safety and efficacy trial. Here, we present the available data, up to 14 days after a second dose in adults (18–55 years of age) from an ongoing phase I/II vaccine study with BNT162b1, which is also enrolling adults who are 65–85 years of age (ClinicalTrials.gov identifier, NCT04368728).

Study design and demographics

Between 4 May 2020 and 19 June 2020, 76 participants were screened, and 45 participants were randomized and vaccinated. Per dose level (10 µg and 30 µg), 12 participants were vaccinated with BNT162b1 on days 1 and 21, 12 participants received a 100-µg dose on day 1 and 9 participants received placebo (Fig. 1). The study population consisted of healthy male and female participants with a mean age of 35.4 years (range, 19–54 years); 51.1% were male and 48.9% were female. Most participants self-reported as white (82.2%) and non-Hispanic/non-Latinx (93.3%) (Extended Data Table 1).

Safety and tolerability

In the 7 days after vaccination doses 1 and 2, pain at the injection site was the most-frequent solicited local reaction, reported after the first dose by 58.3% (7 out of 12) in the 10-µg BNT162b1 group, 100.0% (12 out of 12 each) in the 30-µg and 100-µg BNT162b1 groups, and 22.2% (2 out of 9) in the placebo group. After the second dose, pain was reported by 83.3% (10 out of 12) and 100.0% of individuals who received 10 µg and 30 µg BNT162b1, respectively, and by 16.7% of individuals who received the placebo. All local reactions were mild or moderate in severity except for one report of severe pain after the first dose of 100 µg BNT162b1 (Fig. 2).

The most-common systemic events reported in the 7 days after each vaccination in both BNT162b1 and placebo groups were mild to moderate fatigue and headache. Reports of fatigue and headache were more common in the BNT162b1 groups than in the placebo group. In addition, chills, muscle pain and joint pain were reported by individuals who received BNT162b1 but not by individuals who received the placebo. Systemic events increased with dose level and were reported in a greater number of participants after the second dose (10-µg and 30-µg groups). After the first dose, fever (defined as $\geq 38.0^\circ\text{C}$) was reported by 8.3% (1 out of 12) of participants who received 10 µg and 30 µg BNT162b1 and by 50.0% (6 out of 12) of individuals who received 100 µg BNT162b1. After the second dose, 8.3% (1 out of 12) of participants who received 10 µg BNT162b1 and 75.0% (9 out of 12) of participants who received 30 µg BNT162b1 reported fever of $\geq 38.0^\circ\text{C}$. On the basis of the reactivity reported after the first dose of 100 µg and the second dose of

30 µg, participants who received an initial 100-µg dose did not receive a second 100-µg dose. Fevers generally resolved within 1 day of onset. No grade 4 systemic events or fever were reported (Fig. 3a, b). Most local reactions and systemic events peaked by day 2 after vaccination and resolved by day 7.

Adverse events (Extended Data Table 2) were reported by 50.0% (6 out of 12) of participants who received either 10 or 30 µg of BNT162b1, 58.3% (7 out of 12) of participants who received 100 µg of BNT162b1, and 11.1% (1 out of 9) of placebo recipients. Two participants reported a severe adverse event: grade 3 fever 2 days after vaccination in the 30-µg group, and sleep disturbance 1 day after vaccination in the 100-µg group. Related adverse events were reported by 25% (3 out of 12 in the 10-µg group) to 50% (6 out of 12 each in the 30-µg and 100-µg groups) of individuals who received BNT162b1 and by 11.1% (1 out of 9) of participants who received the placebo. No serious adverse events were reported.

No grade 1 or greater change in routine clinical laboratory values or laboratory abnormalities were observed for most participants after either of the BNT162b1 vaccinations. Of those with laboratory changes, the largest changes were decreases in the lymphocyte count after the first dose in 8.3% (1 out of 12), 45.5% (5 out of 11) and 50.0% (6 out of 12) of participants who received 10 µg, 30 µg and 100 µg BNT162b1, respectively. One participant each in the 10-µg (8.3% (1 out of 12)) and 30-µg (9.1% (1 out of 11)) groups and 4 participants in the 100-µg group (33.3% (4 out of 12)) had grade 3 decreases in the lymphocyte count. These decreases in lymphocyte count after the first dose were transient and returned to normal 6–8 days after vaccination (Extended Data Fig. 1). In addition, grade-2 neutropenia was noted 6–8 days after the second dose in 1 participant each in the 10-µg and 30-µg BNT162b1 groups. These two participants continue to be followed in the study, and no adverse events or clinical manifestations of neutropenia have been reported to date. None of the post-vaccination abnormalities observed were associated with clinical findings.

Immunogenicity

RBD-binding IgG concentrations and SARS-CoV-2-neutralizing titres were assessed at baseline, at 7 and 21 days after the first dose, at 7 days (day 28) and 14 days (day 35) after the second dose of BNT162b1. By 21 days after the first dose (for all three dose levels), geometric mean concentrations (GMCs) of RBD-binding IgG ranged from 534 to 1,778 U ml⁻¹ (Fig. 4a). In comparison, a panel of 38 SARS-CoV-2 infection and/or COVID-19 convalescent sera drawn at least 14 days after a PCR-confirmed diagnosis from patients with COVID-19 (18–83 years

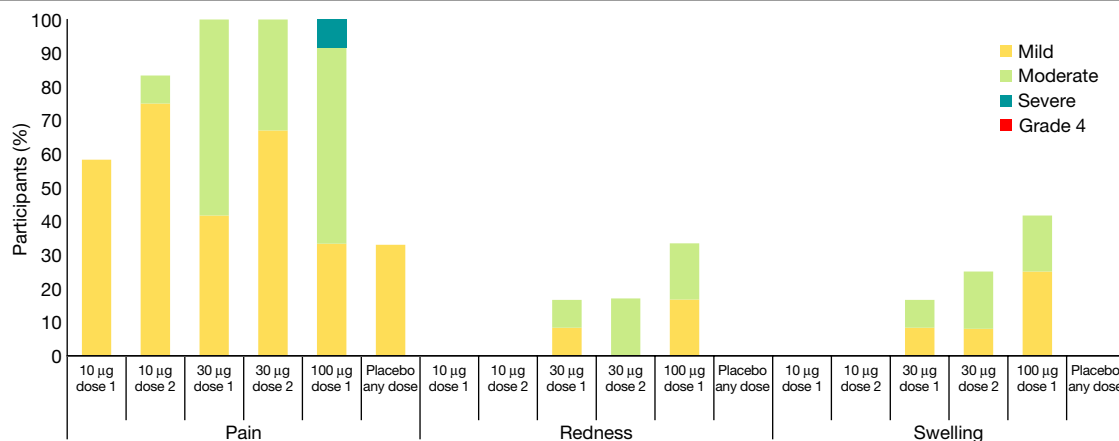


Fig. 2 | Local reactions reported within 7 days of vaccination for all dose levels. Solicited injection-site (local) reactions were: pain at injection site (mild, does not interfere with activity; moderate, interferes with activity; severe, prevents daily activity; grade 4, emergency room visit or hospitalization) and redness and swelling (mild, 2.0–5.0 cm in diameter;

moderate, >5.0–10.0 cm in diameter; severe, >10.0 cm in diameter; grade 4: necrosis or exfoliative dermatitis for redness, and necrosis for swelling). Data were collected with the use of electronic diaries for 7 days after each vaccination.

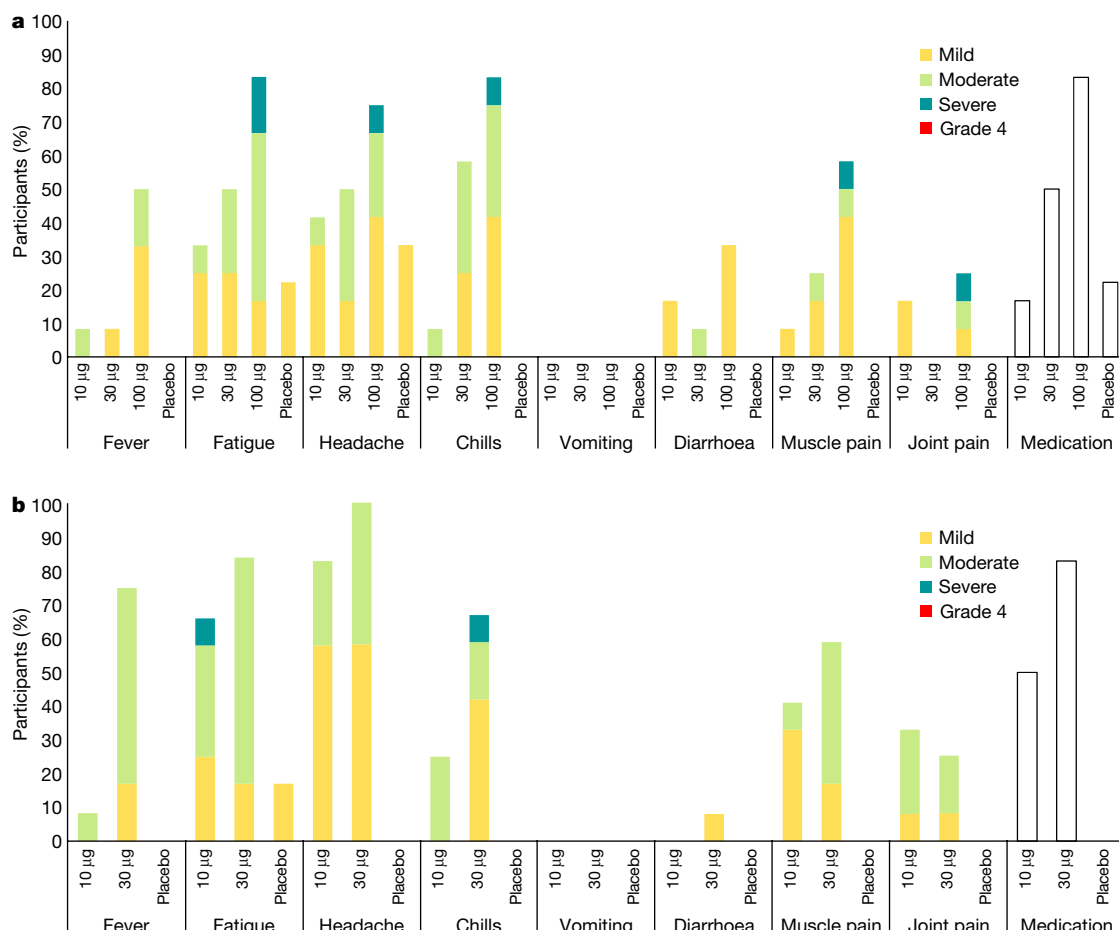


Fig. 3 | Systemic events and medication use reported within 7 days after vaccination. a. Systemic events and medication use reported within 7 days after vaccination 1 for all dose levels. **b.** Systemic events and medication use reported within 7 days after vaccination 2 for the 10-µg and 30-µg dose levels. Solicited systemic events were: fatigue, headache, chills, new or worsened muscle pain, new or worsened joint pain (mild, does not interfere with activity; moderate, some interference with activity; severe, prevents daily activity), vomiting (mild, 1–2 times in 24 h; moderate, >2 times in 24 h; severe, requires

intravenous hydration), diarrhoea (mild, 2–3 loose stools in 24 h; moderate, 4–5 loose stools in 24 h; severe, 6 or more loose stools in 24 h); grade 4 for all events: emergency room visit or hospitalization; and fever (mild, 38.0–38.4 °C; moderate, 38.5–38.9 °C; severe, 39.0–40.0 °C; grade 4, >40.0 °C). Medication indicates the proportion of participants who reported the use of antipyretic or pain medication. Data were collected with the use of electronic diaries for 7 days after each vaccination.

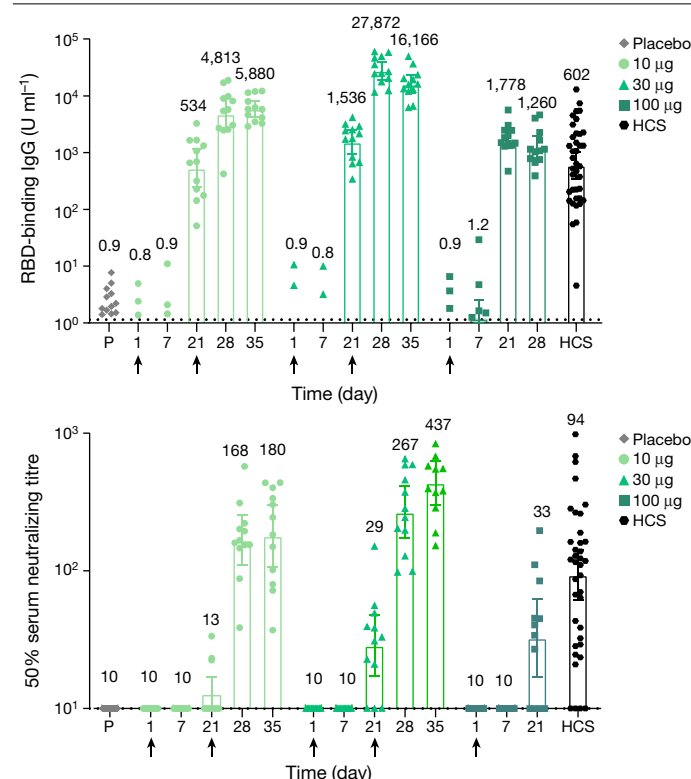


Fig. 4 | Immunogenicity of BNT162b1. Participants in groups of 15 were vaccinated with the indicated dose levels of BNT162b1 ($n = 12$) or with placebo ($n = 3$) on days 1 (all dose levels and placebo) and 21 (10- μ g and 30- μ g dose levels and placebo). Responses in individuals who received the placebo for each of the dosing groups are combined. The 28- and 35-day blood samples were obtained 7 and 14 days after the second vaccination. Sera were obtained before vaccination (day 1), and 7, 21, 28 and 35 days after the first vaccination. Human COVID-19 convalescent sera (HCS, $n = 38$) were obtained at least 14 days after PCR-confirmed diagnosis and at a time when the donors were asymptomatic. **a**, GMCs of recombinant RBD-binding IgG. Because the measured antibody concentrations using the Luminex assay are obtained in arbitrary units, they cannot be directly translated into concentrations on a molar or mass basis. The lower limit of quantitation is 1.15. **b**, The 50% SARS-CoV-2-neutralizing GMTs. Each data point represents a serum sample, and each vertical bar represents a geometric mean with 95% confidence interval. The number above the bars are either the GMC (**a**) or GMT (**b**) for the group. Arrows indicate the timing of vaccination (blood was obtained before vaccination on the vaccination days).

of age) had an RBD-binding IgG GMC of 602 U ml⁻¹. (Additional information on the convalescent serum panel is included in the Methods.) By 7 days after the second dose (for the 10- μ g and 30- μ g dose levels), RBD-binding IgG GMCs had increased to 4,813 and to 27,872 U ml⁻¹, respectively. RBD-binding antibody concentrations among participants who received one dose of 100 μ g BNT162b1 did not increase further at 21 days after the first vaccination. In the participants who received the 10- μ g and 30- μ g doses of BNT162b1, highly elevated RBD-binding antibody concentrations persisted to the last time point evaluated (day 35, 14 days after the second dose). These RBD-binding antibody concentrations were 5,880–16,166 U ml⁻¹ compared to 602 U ml⁻¹ in the panel of human convalescent sera.

For all doses, small increases in SARS-CoV-2-neutralizing geometric mean titres (GMTs) were observed 21 days after the first dose (Fig. 4b). Substantially greater serum neutralizing GMTs were achieved 7 days after the second 10- μ g and 30- μ g dose, reaching 168–267. Neutralizing GMTs further increased by 14 days after the second dose to 180 (10- μ g dose level) and 437 (30- μ g dose level), compared to 94 for the panel of human convalescent sera. The kinetics and durability of the neutralizing titres are being monitored.

Discussion

The RNA-based SARS-CoV-2 vaccine candidate BNT162b1, which was administered as 10- μ g, 30- μ g or 100- μ g doses in healthy adults (18–55 years of age), exhibited a tolerability and safety profile consistent with those previously observed for mRNA-based vaccines⁵. A clear dose-level response in elicited neutralizing titres was observed after doses 1 and 2 in participants with a particularly steep dose response between the 10 μ g and 30 μ g dose levels.

On the basis of the tolerability profile of the first dose at 100 μ g and the second dose at 30 μ g, participants randomized to the 100- μ g group did not receive a second vaccination. Reactogenicity was generally greater after the second dose in the other two dosing levels; however, symptoms were transient and resolved within a few days. Transient decreases in lymphocyte counts (grades 1–3) were observed within a few days after vaccination, and returned to baseline within 6–8 days in all participants. These laboratory abnormalities were not associated with clinical findings. RNA vaccines are known to induce type-I interferon, which has been associated with transient migration of lymphocytes into tissues^{19–22}.

Robust immunogenicity was observed after vaccination with BNT162b1. RBD-binding IgG concentrations were detected at 21 days after the first dose, and these were substantially increased 7 days after the second dose given at day 21. After the first dose, the RBD-binding IgG GMCs (10- μ g dose) were similar to those observed in a panel of 38 convalescent human serum samples, obtained at least 14 days after a PCR-confirmed diagnosis of SARS-CoV-2 infection and/or COVID-19. After the first dose, GMCs were similar in the 30- μ g and 100- μ g groups and higher than those in the panel of human convalescent sera. After the second dose, with 10 μ g or 30 μ g BNT162b1, the RBD-binding IgG GMCs were around 8.0–50-fold that of the GMC of the convalescent serum panel.

The higher RBD-binding IgG GMC elicited by the vaccine relative to the GMC of the human convalescent serum panel may be attributed, in part, to antibodies that bind to epitopes that are exposed on the RNA-expressed RBD immunogen and the recombinant RBD target antigen of the binding assay but are buried and inaccessible to antibodies on the RBDs that are incorporated into the spikes of SARS-CoV-2 virions. Neutralization provides a measure of the vaccine-elicited antibody response that is more relevant to potential protection. Neutralization titres were measurable after a single vaccination at day 21 for all dose levels. At day 28 (7 days after the second dose), substantial SARS-CoV-2 neutralization titres were observed. The virus-neutralizing GMTs after the second dose of 10 μ g and 30 μ g were, respectively, 1.8-fold and 2.8-fold the GMT of the convalescent serum panel. By day 35 (14 days after the second dose)—despite a decrease in RBD-binding IgG titres since day 28—neutralizing GMTs continued to increase, to 1.9-fold and 4.6-fold the GMT of the convalescent panel for the 10 μ g and 30 μ g doses, respectively, which is consistent with affinity maturation.

Assuming that the neutralization titres that are induced by natural infection provide protection from COVID-19, comparing vaccine-induced SARS-CoV-2 neutralization titres to those from sera of convalescent humans provides a benchmark for the magnitude of the vaccine-elicited response and the potential of the vaccine to provide protection. Because the titre at which human neutralizing antibodies are protective remains unknown, these findings are not proof of vaccine efficacy. Efficacy will be determined in a pivotal phase III trial. Because the cohort that received the 100 μ g dose level did not receive the booster dose, no data for immunogenicity after a second vaccination at this dose level are available; however, there were no substantial differences in immunogenicity between the 30- μ g and 100- μ g dose levels after the first dose. This observation suggests that a well-tolerated and immunogenic dose level may be between 10 μ g and 30 μ g for this vaccine candidate.

Our study had several limitations. Although we used convalescent sera as a comparator, the kind of immunity (T cells versus B cells or both) and level of immunity needed to protect from COVID-19 are unknown. Furthermore, this analysis of available data did not assess

immune responses or safety beyond 2 weeks after the second dose of vaccine. Both are important to inform the public health use of this vaccine. Follow-up will continue for all participants and will include collection of serious adverse events for 6 months and COVID-19 infection and multiple additional immunogenicity measurements for up to 2 years. Although our population of healthy adults up to 55 years of age is appropriate for a phase I/II study, it does not accurately reflect the population at highest risk for COVID-19. Adults who are 65 years of age and over have already been enrolled in this study and results will be reported as they become available. Later phases of this study will prioritize enrolment of more diverse populations, including those with chronic underlying health conditions and from racial and ethnic groups that are adversely affected by COVID-19²³.

The clinical testing of BNT162b1 described here has taken place in the context of a broader, ongoing COVID-19-vaccine-development program. That program includes the clinical testing of three additional vaccine candidates, including candidates that encode the full-length spike protein, and a parallel trial in Germany, in which additional immune responses, including neutralizing responses against variant strains and cell-mediated responses, are being assessed²⁴. The resulting comparative data will allow us to address whether a full-length spike immunogen, which presents additional epitopes, is better able to elicit high virus-neutralizing titres that are robust to potential antigenic drift of SARS-CoV-2 than the relatively small RBD immunogen that is encoded by BNT162b1. The clinical findings for the BNT162b1 RNA-based vaccine candidate are encouraging and strongly support accelerated clinical development, including efficacy testing, and at-risk manufacturing to maximize the opportunity for the rapid production of a SARS-CoV-2 vaccine to prevent COVID-19.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-020-2639-4>.

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Study design

This study was conducted in healthy men and women (who were not pregnant) who were 18–55 years of age to assess the safety, tolerability and immunogenicity of ascending dose levels of various BNT162 mRNA vaccine candidates. In the part of the study reported here, assessment of three dose levels (10 µg, 30 µg or 100 µg) of the BNT162b1 candidate was conducted at two sites in the USA. This study used a sentinel cohort design with progression and dose escalation taking place after review of data from the sentinel cohort at each dose level. The study is registered at ClinicalTrials.gov (NCT04368728). The phase I portion of this study was observer-blinded at the site level. Investigators were blinded to participant-level study intervention assignment; but investigators were not blinded to group-level assignment for the dataset included in this Article.

Eligibility

Key exclusion criteria included individuals with known infection with human immunodeficiency virus, hepatitis C virus or hepatitis B virus; immunocompromised individuals and those with a history of autoimmune disease; and those with increased risk for severe COVID-19, previous clinical or microbiological diagnosis of COVID-19, receipt of medications intended to prevent COVID-19, previous vaccination with any coronavirus vaccine, a positive serological test for SARS-CoV-2 IgM and/or IgG at the screening visit, and a SARS-CoV-2 nucleic acid amplification test-positive nasal swab within 24 h before study vaccination.

The final protocol and informed consent document were approved by institutional review boards for each of the participating investigational centres. This study was conducted in compliance with all International Council for Harmonisation good clinical practice guidelines and the ethical principles of the Declaration of Helsinki. A signed and dated informed consent form was required before any study-specific activity was performed.

End points

In this report, results from the following study primary end points are presented: the proportion of participants who reported solicited local reactions, systemic events and use of antipyretic and/or pain medication within 7 days after vaccination, adverse events and serious adverse events (available up to around 45 days after dose 1), and the proportion of participants with clinical laboratory abnormalities 1 and 7 days after vaccination and grading shifts in laboratory assessments between baseline and 1 and 7 days after dose 1, and between dose 2 and 7 days after dose 2. Secondary end points included: SARS-CoV-2-neutralizing GMTs and SARS-CoV-2 RBD-binding IgG GMCs 7 and 21 days after dose 1, and 7 and 14 days after dose 2.

Procedures

Study participants were randomly assigned to a vaccine group using an interactive web-based response technology system with each group comprising 15 participants (12 active vaccine recipients and 3 placebo recipients). Participants received two 0.5-ml doses of either BNT162b1 or placebo, administered by intramuscular injection into the deltoid muscle.

BNT162b1 incorporates a good manufacturing practice-grade mRNA drug substance that encodes the trimerized SARS-CoV-2 spike glycoprotein RBD antigen. The coding sequence for the antigen has been deposited with GenBank (accession number, MN908947.3). The mRNA is formulated with lipids as the mRNA–lipid nanoparticle drug product. The vaccine was supplied as a buffered-liquid solution for intramuscular injection and was stored at –80 °C. The placebo was a sterile saline solution for injection (0.9% sodium chloride injection, in a 0.5-ml dose).

Safety assessments

Safety assessments included a 4-h observation after vaccination (for the first 5 participants vaccinated in each group), or a 30-min observation (for the remainder of participants) for immediate adverse events. The safety assessments also included self-reporting of solicited local reactions (redness, swelling and pain at the injection site), systemic events (fever, fatigue, headache, chills, vomiting, diarrhoea, muscle pain and joint pain), the use of antipyretic and/or pain medication in an electronic diary for 7 days after vaccination, and the reporting of unsolicited adverse events and serious adverse events after vaccination. Haematology and chemistry assessments were conducted at screening, 1 and 7 days after the first dose, and 7 days after the second dose.

There were protocol-specified safety stopping rules for all sentinel cohort participants. Both an internal review committee and an external data monitoring committee reviewed all safety data. No stopping rules were met before the publication of this report.

Human convalescent serum panel

The 38 human SARS-CoV-2 infection and/or COVID-19 convalescent sera were drawn from participants, who were 18–83 years of age, at least 14 days after PCR-confirmed diagnosis, and at a time when participants were asymptomatic. The mean age of the donors was 45 years of age. Neutralizing GMTs in subgroups of the donors were as follows: ≤55 years of age, 82 (*n* = 29); >55 years of age, 142 (*n* = 9); symptomatic infections, 90 (*n* = 35); asymptomatic infections, 156 (*n* = 3). The antibody titre for the one individual who was hospitalized was 618. The sera were obtained from Sanguine Biosciences, the MT Group and Pfizer Occupational Health and Wellness.

Immunogenicity assessments

For immunogenicity assessments, 50 ml of blood was collected before each study vaccination, at 7 and 21 days after the first dose, and at 7 and 14 days after the second dose. In the RBD-binding IgG assay, a recombinant SARS-CoV-2 RBD containing a C-terminal Avitag (Acro Biosystems, SPD-C82E9) and no foldon domain was bound to streptavidin-coated Luminex microspheres. In brief, 1.25×10^7 microspheres/ml were coated with streptavidin by 1-ethyl-3-[3-dimethylaminopropyl] carbodiimide hydrochloride reaction. Recombinant RBD Avitag was coupled to streptavidin beads by incubating for 90 min at room temperature with shaking (35 rpm). Beads were blocked in 1% BSA buffer for 30 min at room temperature. Heat-inactivated serum from participants was diluted 1:500, 1:5,000 and 1:50,000 in assay buffer (PBS with 0.5% BSA, 0.05% Tween-20 and 0.02% sodium azide). Following a 16–20-h incubation at 2–8 °C with shaking (300 rpm), plates were washed three times in a solution containing 0.05% Tween-20. An R-phycoerythrin-conjugated goat anti-human polyclonal antibody (Jackson Labs) was then added to plates for 90 min at room temperature with shaking (300 RPM). Plates were then washed a final time in a solution containing 0.05% Tween-20. Data were captured as median fluorescent intensities using a Luminex reader and converted to U/ml antibody concentrations using a reference standard curve with arbitrary assigned concentrations of 100 U/ml and accounting for the serum dilution factor. The reference standard was composed of a pool of five COVID-19 convalescent serum samples (>14 days after PCR diagnosis). Three dilutions are used to increase the likelihood that at least one result for any sample will fall within the usable range of the standard curve. Assay results were reported in U/ml of IgG. The final assay results are expressed as the GMC of all sample dilutions that produced a valid assay result within the assay range.

The SARS-CoV-2 neutralization assay used a previously described strain of SARS-CoV-2 (USA_WA1/2020) that had been rescued by reverse genetics and engineered by the insertion of an mNeonGreen gene into open-reading frame 7 of the viral genome²⁵. This reporter virus generates similar plaque morphologies and indistinguishable growth curves from the wild-type virus. Viral master stocks (2×10^7

plaque-forming units per ml) used for the neutralization assay were grown in Vero E6 cells as previously described²⁵. When testing patient convalescent serum specimens, the fluorescent neutralization assay produced comparable results as the conventional plaque reduction neutralization assay²⁶. In brief, serial dilutions of heat-inactivated sera from participants were incubated with the reporter virus to yield an infection rate of approximately 10–30% of the Vero monolayer for 1 h at 37 °C before inoculating Vero CCL81 cell monolayers (targeted to have 8,000–15,000 cells per well) in 96-well plates to enable the accurate quantification of infected cells. Total cell counts per well were enumerated by nuclear stain (Hoechst 33342) and fluorescent virally infected foci were detected 16–24 h after inoculation with a Cytation 7 Cell Imaging Multi-Mode Reader (BioTek) with Gen5 Image Prime v.3.09. Titres were calculated in GraphPad Prism v.8.4.2 by generating a four-parameter logistical fit of the percentage neutralization at each serial serum dilution. The 50% neutralization titre was reported as the interpolated reciprocal of the dilution that yielded a 50% reduction in fluorescent viral foci.

Statistical analysis

The sample size for the reported part of the study was not based on statistical hypothesis testing. The primary safety objective was evaluated by descriptive summary statistics for local reactions, systemic events, abnormal haematology and chemistry laboratory parameters, adverse events and serious adverse events after each vaccine dose for each vaccine group. The secondary immunogenicity objectives were descriptively summarized at the various time points. All participants with data available were included in the safety and immunogenicity analyses.

Reporting summary

Further information on research design is available in the Nature Research Reporting Summary linked to this paper.

Data availability

Upon request, and subject to review, Pfizer will provide the data that support the findings of this study. Subject to certain criteria, conditions and exceptions, Pfizer may also provide access to the related individual anonymized participant data. See <https://www.pfizer.com/science/clinical-trials/trial-data-and-results> for more information. These data

are interim data from an ongoing study for which the database is not locked. Data have not yet been source-verified or subjected to standard quality check procedures that would occur at the time of database lock and may therefore be subject to change.

25. Xie, X. et al. An infectious cDNA clone of SARS-CoV-2. *Cell Host Microbe* **27**, 841–848 (2020).
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Author contributions K.U.J., P.R.D., W.C.G., N.K., S.L., A.G., R.B., Ö.T. and U.S. were involved in the design of the overall study and strategy. K.N., M.J.M., E.E.W., R.F. and A.R.F. provided feedback on the study design. W.K., D.C., K.A.S., K.R.T., C.F.-G. and P.-Y.S. performed the immunological analyses. M.J.M., K.N., E.E.W., R.F., A.R.F., K.E.L. and V.R. collected data as study investigators. P.L. and K.K. developed the statistical design and oversaw the data analysis. J.A., K.U.J., P.R.D. and W.C.G. drafted the initial version of the manuscript. All authors reviewed and edited the manuscript and approved the final version.

Competing interests N.K., J.A., A.G., S.L., R.B., K.A.S., P.L., K.K., W.K., D.C., K.R.T., P.R.D., W.C.G. and K.U.J. are employees of Pfizer and may hold stock options. U.S. and Ö.T. are stock owners, management board members and employees at BioNTech and are inventors on patents and patent applications related to RNA technology. M.J.M., K.E.L., K.N., E.E.W., A.R.F., R.F. and V.R. received compensation from Pfizer for their role as study investigators. C.F.-G. and P.-Y.S. received compensation from Pfizer to perform the neutralization assay.

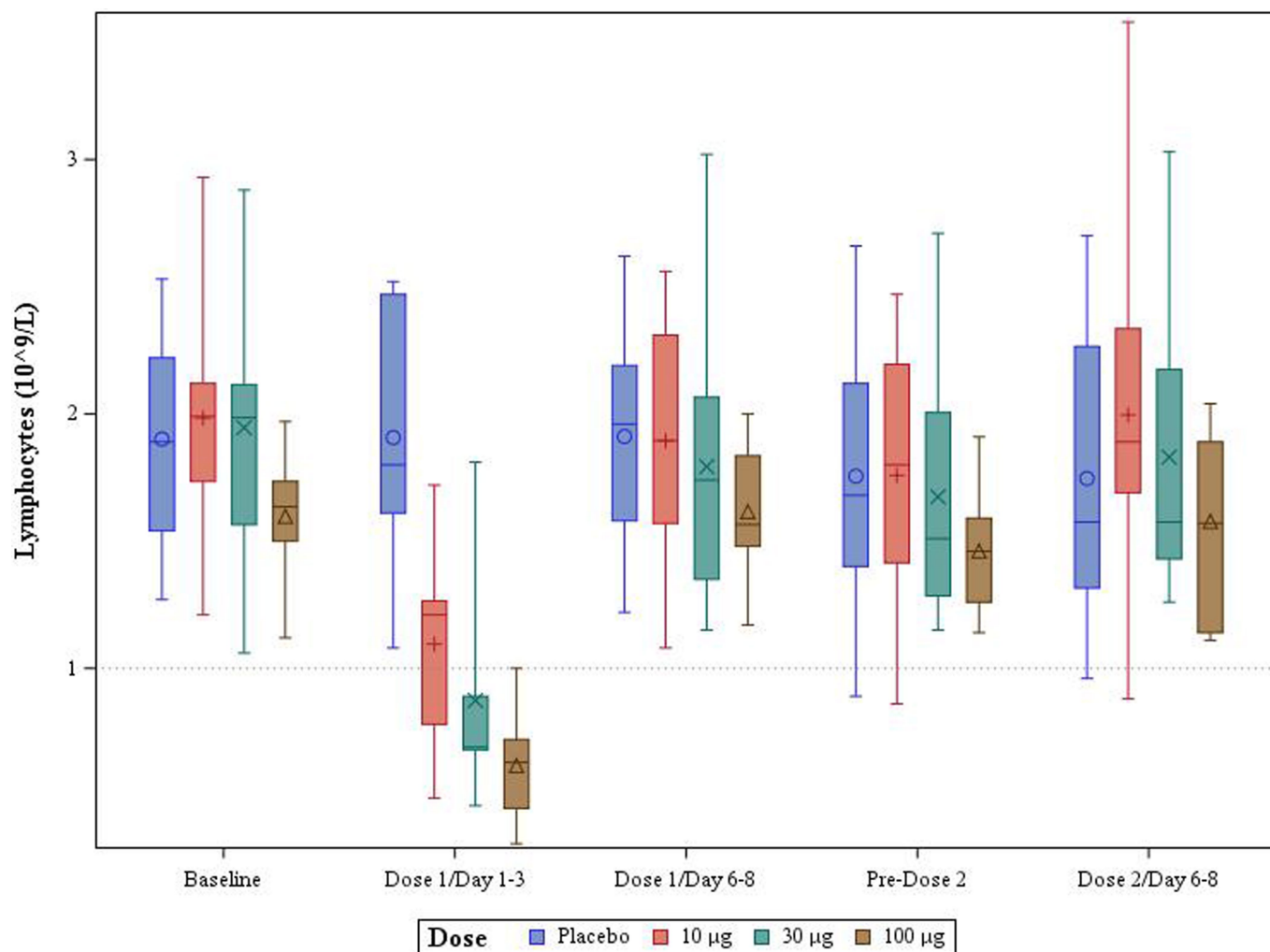
Additional information

Supplementary information is available for this paper at <https://doi.org/10.1038/s41586-020-2639-4>.

Correspondence and requests for materials should be addressed to J.A.

Peer review information Nature thanks Barbra Richardson and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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Extended Data Fig. 1 | Post vaccination changes in lymphocyte count over time. The following time points are shown: dose 1/day 1-3, around 1 day after dose 1; dose 1/day 6-8, around 7 days after dose 1; pre-dose 2, before dose 2; dose 2/day 6-8, around 7 days after dose 2. Symbols denote group means;

circle, placebo; plus, 10 µg; cross, 30 µg; triangle, 100 µg. The box-and-whisker plots show the median (centre), first and third quartiles (lower and upper edges), and minimum and maximum values (lower and upper whiskers).

Extended Data Table 1 | Demographic characteristics

	10 µg (N=12) n (%)	30 µg (N=12) n (%)	100 µg (N=12) n (%)	Placebo (N=9) n (%)	Total (N=45) n (%)
Sex					
Male	7 (58.3)	6 (50.0)	5 (41.7)	5 (55.6)	23 (51.1)
Female	5 (41.7)	6 (50.0)	7 (58.3)	4 (44.4)	22 (48.9)
Race					
White	8 (66.7)	10 (83.3)	11 (91.7)	8 (88.9)	37 (82.2)
Black or African American	1 (8.3)	0	0	0	1 (2.2)
Asian	3 (25.0)	2 (16.7)	1 (8.3)	1 (11.1)	7 (15.6)
Ethnicity					
Hispanic/Latino	1 (8.3)	1 (8.3)	0	0	2 (4.4)
Non-Hispanic/non-Latino	11 (91.7)	10 (83.3)	12 (100.0)	9 (100.0)	42 (93.3)
Not reported	0	1 (8.3)	0	0	1 (2.2)
Age at vaccination (years)					
Mean (SD)	29.4 (6.39)	35.8 (9.96)	38.3 (9.34)	39.0 (11.16)	35.4 (9.71)
Median	26.5	33.5	38.0	41.0	33.0
Min, max	(24, 42)	(23, 52)	(25, 53)	(19, 54)	(19, 54)

N, the number of participants in the specified group or the total sample. This value is the denominator for the percentage calculations. n, the number of participants with the specified characteristic.

Extended Data Table 2 | Adverse events

	10 µg (N=12)	30 µg (N=12)	100 µg (N=12)	Placebo (N=9)
Adverse Event	n (%)	n (%)	n (%)	n (%)
Any event	6 (50.0)	6 (50.0)	7 (58.3)	1 (11.1)
Related	3 (25.0)	6 (50.0)	6 (50.0)	1 (11.1)
Severe	0	1 (8.3)	1 (8.3)	0
Life-threatening	0	0	0	0
Any serious adverse event	0	0	0	0
Related	0	0	0	0
Severe	0	0	0	0
Life-threatening	0	0	0	0
Any adverse event leading to withdrawal	0	0	0	0
Related	0	0	0	0
Severe	0	0	0	0
Life-threatening	0	0	0	0
Death	0	0	0	0

N, the number of participants in the specified group or the total sample. This value is the denominator for the percentage calculations. n, the number of participants who reported at least one occurrence of the specified adverse event category. For 'any event', n indicates the number of participants who reported at least one occurrence of any adverse event. Related, assessed by the investigator as related to the investigational product.

Reporting Summary

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For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

- | | |
|-------------------------------------|--|
| n/a | Confirmed |
| ☐ | ☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement |
| ☐ | ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly |
| ☐ | ☒ The statistical test(s) used AND whether they are one or two sided
<i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i> |
| ☒ | ☐ A description of all covariates tested |
| ☒ | ☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons |
| ☐ | ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals) |
| ☒ | ☐ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted
<i>Give P values as exact values whenever suitable.</i> |
| ☒ | ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☒ | ☐ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated |

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

- | | |
|-----------------|---|
| Data collection | Inform (for data collected in the case report form) and electronic diary (Signant Health platform) for participant self reported reactogenicity |
| Data analysis | SAS 9.4 |

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research [guidelines for submitting code & software](#) for further information.

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	The sample size for this interim report was not based on statistical hypothesis testing. A total of 45 participants were enrolled in this part of the study. For the purposes of tolerability and dose escalation study a total of 15 participants (12 receiving vaccine and 3 receiving placebo) was deemed sufficient for a dosing finding phase study.
Data exclusions	All safety and immunogenicity data that were available at the time of the data snapshot were included in the interim report. No data were excluded from the analyses.
Replication	This is an interim report of an ongoing human clinical trial. There was no attempt at replication of study findings
Randomization	This is an randomized controlled trial. Study participants were randomly assigned to a vaccine group using an interactive web based response technology system with each group comprising 15 participants (12 active vaccine recipients and 3 placebo recipients).
Blinding	This is an observer blinded study which is investigator blinded but Sponsor unblinded during Stage 1 (the stage from which data in the manuscript are presented). Investigators were unblinded to group level data but not subject level data for the purposes of interpretation and summary of the results included in this interim report.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☐	☒ Human research participants
☐	☒ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP seq
☒	☐ Flow cytometry
☒	☐ MRI based neuroimaging

Human research participants

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Population characteristics	Study participants were healthy men or women 18-55 years of age. Key exclusion criteria included individuals with known infection with human immunodeficiency virus, hepatitis C virus, or hepatitis B virus; immunocompromised individuals and those with a history of autoimmune disease; those with increased risk for severe COVID-19; previous clinical or microbiological diagnosis of COVID-19; receipt of medications intended to prevent COVID-19; previous vaccination with any coronavirus vaccine; a positive serological test for SARS-CoV-2 IgM and/or IgG at the screening visit; and a SARS-CoV-2 NAAT positive nasal swab within 24 hours before study vaccination.
Recruitment	Study participants were recruited at the two individual sites and recruitment strategies were at the discretion of individual sites and could include identification of interested individuals from the sites local database or through advertising in the local community. Once recruited participants were screened for eligibility based on pre-specified protocol criteria. Eligible participants were then randomized to vaccine or placebo in a blinded manner. These processes therefore did not lead themselves to enrollment biases however participants who did not know about the study may have had less of an opportunity to participate.
Ethics oversight	The study protocol was approved by the western institutional review board for one site and by the Langone Health New York University Institutional IRB prior to enrollment of any participants

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Clinical data

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Clinical trial registration	ClinicalTrials.gov identifier: NCT04368728
Study protocol	Details of protocol elements can be accessed from clinicaltrials.gov
Data collection	Data were collected at screening (up to 14 days before vaccination) and for randomized participants at the investigative site at baseline, 1 day, 7 days and 21 days, after Dose 1, 7 days after dose 2 and up to 14 days after dose 2. Both safety and/or serum collection for immunogenicity assessments were collected for all stated time points. In addition, reactogenicity data were assessed through participant self reports via an electronic diary for 7 days after dose 1.
Outcomes	In this interim report, the following study primary endpoints are presented: the proportion of participants reporting prompted local reactions, systemic events, and use of antipyretic and/or pain medication within 7 days after vaccination, AEs and serious adverse events (SAEs) (available through up to ~45 days after Dose 1), and the proportion of participants with clinical laboratory abnormalities 1 and 7 days after vaccination and grading shifts in laboratory assessments between baseline and 1 and 7 days after Dose 1 and between Dose 2 and 7 days after Dose 2. Secondary endpoints included: SARS CoV 2 neutralizing geometric mean titers (GMTs); SARS CoV 2 RBD binding IgG geometric mean concentrations (GMCs) 7 and 21 days after Dose 1 and 7 and 14 days after Dose 2

ORIGINAL ARTICLE

Safety and Immunogenicity of Two RNA-Based Covid-19 Vaccine Candidates

Edward E. Walsh, M.D., Robert W. Frenck, Jr., M.D., Ann R. Falsey, M.D., Nicholas Kitchin, M.D., Judith Absalon, M.D., Alejandra Gurtman, M.D., Stephen Lockhart, D.M., Kathleen Neuzil, M.D., Mark J. Mulligan, M.D., Ruth Bailey, B.Sc., Kena A. Swanson, Ph.D., Ping Li, Ph.D., Kenneth Koury, Ph.D., Warren Kalina, Ph.D., David Cooper, Ph.D., Camila Fontes-Garfias, B.Sc., Pei-Yong Shi, Ph.D., Özlem Türeci, M.D., Kristin R. Tompkins, B.Sc., Kirsten E. Lyke, M.D., Vanessa Raabe, M.D., Philip R. Dormitzer, M.D., Kathrin U. Jansen, Ph.D., Uğur Şahin, M.D., and William C. Gruber, M.D.

ABSTRACT

BACKGROUND

Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infections and the resulting disease, coronavirus disease 2019 (Covid-19), have spread to millions of persons worldwide. Multiple vaccine candidates are under development, but no vaccine is currently available. Interim safety and immunogenicity data about the vaccine candidate BNT162b1 in younger adults have been reported previously from trials in Germany and the United States.

METHODS

In an ongoing, placebo-controlled, observer-blinded, dose-escalation, phase 1 trial conducted in the United States, we randomly assigned healthy adults 18 to 55 years of age and those 65 to 85 years of age to receive either placebo or one of two lipid nanoparticle–formulated, nucleoside-modified RNA vaccine candidates: BNT162b1, which encodes a secreted trimerized SARS-CoV-2 receptor–binding domain; or BNT162b2, which encodes a membrane-anchored SARS-CoV-2 full-length spike, stabilized in the prefusion conformation. The primary outcome was safety (e.g., local and systemic reactions and adverse events); immunogenicity was a secondary outcome. Trial groups were defined according to vaccine candidate, age of the participants, and vaccine dose level (10 μ g, 20 μ g, 30 μ g, and 100 μ g). In all groups but one, participants received two doses, with a 21-day interval between doses; in one group (100 μ g of BNT162b1), participants received one dose.

RESULTS

A total of 195 participants underwent randomization. In each of 13 groups of 15 participants, 12 participants received vaccine and 3 received placebo. BNT162b2 was associated with a lower incidence and severity of systemic reactions than BNT162b1, particularly in older adults. In both younger and older adults, the two vaccine candidates elicited similar dose-dependent SARS-CoV-2–neutralizing geometric mean titers, which were similar to or higher than the geometric mean titer of a panel of SARS-CoV-2 convalescent serum samples.

CONCLUSIONS

The safety and immunogenicity data from this U.S. phase 1 trial of two vaccine candidates in younger and older adults, added to earlier interim safety and immunogenicity data regarding BNT162b1 in younger adults from trials in Germany and the United States, support the selection of BNT162b2 for advancement to a pivotal phase 2–3 safety and efficacy evaluation. (Funded by BioNTech and Pfizer; ClinicalTrials.gov number, NCT04368728.)

From the University of Rochester and Rochester General Hospital, Rochester (E.E.W., A.R.F.); Vaccine Research and Development, Pfizer, Pearl River (J.A., A.G., K.A.S., K.K., W.K., D.C., K.R.T., P.R.D., K.U.J., W.C.G.); and New York University Langone Vaccine Center and Grossman School of Medicine, New York (M.J.M., V.R.) — all in New York; Cincinnati Children's Hospital, Cincinnati (R.W.F.); Vaccine Research and Development, Pfizer, Hurley, United Kingdom (N.K., S.L., R.B.); the University of Maryland School of Medicine, Center for Vaccine Development and Global Health, Baltimore (K.N., K.E.L.); Vaccine Research and Development, Pfizer, Collegeville, PA (P.L.); the University of Texas Medical Branch, Galveston (C.F.-G., P.-Y.S.); and BioNTech, Mainz, Germany (Ö.T., U.Ş.). Address reprint requests to Dr. Absalon at Pfizer, 401 N. Middletown Rd., Pearl River, NY 10965, or at judith.absalon@pfizer.com.

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SINCE THE FIRST CASES OF CORONAVIRUS disease 2019 (Covid-19) in Wuhan, China, in December 2019, pandemic illness has spread to millions of persons worldwide. An increased risk of severe disease and death has been noted among the elderly and among persons with preexisting medical conditions. No Covid-19 vaccines are currently available, and they are urgently needed to combat escalating cases and deaths worldwide.¹

In response, BioNTech and Pfizer launched a coordinated program to compare four RNA-based Covid-19 pandemic vaccine candidates in umbrella-type clinical studies conducted in Germany (BNT162-01) and the United States (C4591001). The program was designed to support the selection of a single vaccine candidate and dose level for a pivotal international safety and efficacy trial. On the basis of initial clinical-trial results in Germany,² two lipid nanoparticle–formulated,³ nucleoside-modified RNA (modRNA)⁴ vaccine candidates against severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) were evaluated in the phase 1 portion of the trial in the United States.⁵ One of these candidates, BNT162b1, encodes the SARS-CoV-2 receptor-binding domain, trimerized by the addition of a T4 fibrin foldon domain to increase its immunogenicity through multivalent display.^{6–8} The other candidate, BNT162b2, encodes the SARS-CoV-2 full-length spike, modified by two proline mutations to lock it in the prefusion conformation⁹ and more closely mimic the intact virus with which the elicited virus-neutralizing antibodies must interact.¹⁰

Previous articles have described the assessment of BNT162b1, at multiple dose levels, in healthy adults 18 to 55 years of age.^{2,5} These studies indicated that dose levels of BNT162b1 that elicited an acceptable level of reactogenicity also efficiently elicited titers that were as high as those in a panel of SARS-CoV-2 human convalescent serum samples and that were broadly neutralizing across a panel of 17 SARS-CoV-2 pseudoviruses representing a diversity of circulating strains. BNT162b1 also elicited CD4⁺ type 1 helper T (Th1) cell responses and strong interferon- γ -producing and interleukin-2-producing CD8⁺ cytotoxic T-cell responses. This ability to elicit both humoral and cell-mediated antiviral mechanisms makes BNT162b1 a promising vaccine candidate.

Here, we report the full set of safety and immunogenicity data from the phase 1 portion of an ongoing randomized, placebo-controlled, observer-blinded, dose-escalation trial in the United States that was used to select the final vaccine candidate, as well as the comparison of the safety and immunogenicity of both vaccine candidates and additional phase 1 data that have been collected since candidate selection. These data include evaluation of the 10- μ g, 20- μ g, and 30- μ g dose levels of BNT162b1 and BNT162b2 in adults 18 to 55 years of age and adults 65 to 85 years of age.

METHODS

TRIAL OBJECTIVES, PARTICIPANTS, AND OVERSIGHT

We assessed the safety and immunogenicity of three dose levels of BNT162b1 and BNT162b2. Healthy adults 18 to 55 years of age or 65 to 85 years of age were eligible for inclusion. Key exclusion criteria were known infection with human immunodeficiency virus, hepatitis C virus, or hepatitis B virus; an immunocompromised condition; a history of autoimmune disease; a previous clinical or microbiologic diagnosis of Covid-19; the receipt of medications intended to prevent Covid-19; any previous coronavirus vaccination; positive test for SARS-CoV-2 IgM or IgG at the screening visit; and positive nasal-swab results on a SARS-CoV-2 nucleic acid amplification test within 24 hours before the receipt of trial vaccine or placebo.

BioNTech was the regulatory sponsor of the trial. Pfizer was responsible for the trial design; for the collection, analysis, and interpretation of the data; and for the writing of the report. The corresponding author had full access to all the data in the trial and had final responsibility for the decision to submit the manuscript for publication. All the trial data were available to all the authors.

TRIAL PROCEDURES

Using an interactive Web-based response technology system, we randomly assigned trial participants to groups defined according to the vaccine candidate, dose level, and age range. Groups of participants 18 to 55 years of age and 65 to 85 years of age were to receive doses of 10 μ g, 20 μ g, or 30 μ g of BNT162b1 or BNT162b2 (or placebo) on a two-dose schedule; one group of participants

18 to 55 years of age was assigned to receive 100- μ g doses of BNT162b1 or placebo. All the participants were assigned to receive two 0.5-ml injections of active vaccine (BNT162b1 or BNT162b2) or placebo into the deltoid, administered 21 days apart.

The first five participants in each new dose level or age group (with a randomization ratio of 4:1 for active vaccine:placebo) were observed for 4 hours after the injection to identify immediate adverse events. All the other participants were observed for 30 minutes. Blood samples were obtained for safety and immunogenicity assessments.

SAFETY

The primary end points in phase 1 of this trial were solicited local reactions (i.e., specific local reactions as prompted by and recorded in an electronic diary), systemic events, and use of antipyretic or pain medication within 7 days after the receipt of vaccine or placebo, as prompted by and recorded in an electronic diary; unsolicited adverse events and serious adverse events (i.e., those reported by the participants, without electronic-diary prompts), assessed from the receipt of the first dose through 1 month and 6 months, respectively, after the receipt of the second dose; clinical laboratory abnormalities, assessed 1 day and 7 days after the receipt of vaccine or placebo; and grading shifts in laboratory assessments between baseline and 1 day and 7 days after the first dose and between 2 days and 7 days after the second dose. Protocol-specified safety stopping rules were in effect for all the participants in the phase 1 portion of the trial. The full protocol, including the statistical analysis plan, is available with the full text of this article at NEJM.org. An internal review committee and an external data and safety monitoring committee reviewed all safety data.

IMMUNOGENICITY

Immunogenicity assessments (SARS-CoV-2 serum neutralization assay and receptor-binding domain [RBD]-binding or S1-binding IgG direct Luminex immunoassays) were conducted before the administration of vaccine or placebo, at 7 days and 21 days after the first dose, and at 7 days (i.e., day 28) and 14 days (i.e., day 35) after the second dose. The neutralization assay, which also generated previously described virus-neutralization data from trials of the BNT162 candidates,^{2,5}

used a previously described strain of SARS-CoV-2 (USA_WA1/2020) that had been generated by reverse genetics and engineered by the insertion of an mNeonGreen gene into open reading frame 7 of the viral genome.^{11,12} The 50% neutralization titers and 90% neutralization titers were reported as the interpolated reciprocal of the dilutions yielding 50% and 90% reductions, respectively, in fluorescent viral foci. Any serologic values below the lower limit of quantitation were set to 0.5 times the lower limit of quantitation. Available serologic results were included in the analysis.

Immunogenicity data from a human convalescent serum panel were included as a benchmark. A total of 38 serum samples were obtained from donors 18 to 83 years of age (median age, 42.5 years) who had recovered from SARS-CoV-2 infection or Covid-19; samples were obtained at least 14 days after a polymerase chain reaction-confirmed diagnosis and after symptom resolution. Neutralizing geometric mean titers (GMTs) in subgroups of the donors were as follows: 90, among 35 donors with symptomatic infections; 156, among 3 donors with asymptomatic infection; and 618, in 1 donor who was hospitalized. Each serum sample in the panel was from a different donor. Thus, most of the serum samples were obtained from persons with moderate Covid-19 who had not been hospitalized. The serum samples were obtained from Sanguine Biosciences, the MT Group, and Pfizer Occupational Health and Wellness.

STATISTICAL ANALYSIS

We report descriptive results of safety and immunogenicity analyses, and the sample size was not based on statistical hypothesis testing. Results of the safety analyses are presented as counts, percentages, and associated Clopper-Pearson 95% confidence intervals for local reactions, systemic events, and any adverse events after the administration of vaccine or placebo, according to terms in the *Medical Dictionary for Regulatory Activities*, version 23.0, for each vaccine group. Summary statistics are provided for abnormal laboratory values and grading shifts. Given the small number of participants in each group, the trial was not powered for formal statistical comparisons between dose levels or between age groups.

Immunogenicity analyses of SARS-CoV-2 serum neutralizing titers, S1-binding IgG and RBD-binding IgG concentrations, GMTs, and geometric

mean concentrations (GMCs) were computed along with associated 95% confidence intervals. The GMTs and GMCs were calculated as the mean of the assay results after the logarithmic transformation was made; we then exponentiated the mean to express results on the original scale. Two-sided 95% confidence intervals were obtained by performing logarithmic transformations of titers or concentrations, calculating the 95% confidence interval with reference to Student's *t*-distribution, and then exponentiating the limits of the confidence intervals.

RESULTS

DEMOGRAPHIC CHARACTERISTICS OF THE PARTICIPANTS

Between May 4, 2020, and June 22, 2020, a total of 332 healthy adults (men and nonpregnant women) underwent screening at four sites in the United States (two sites per vaccine candidate). A total of 195 participants were randomly assigned to 13 groups comprising 15 participants each; in each group, 12 participants received vaccine and 3 received placebo (Fig. 1). In all groups

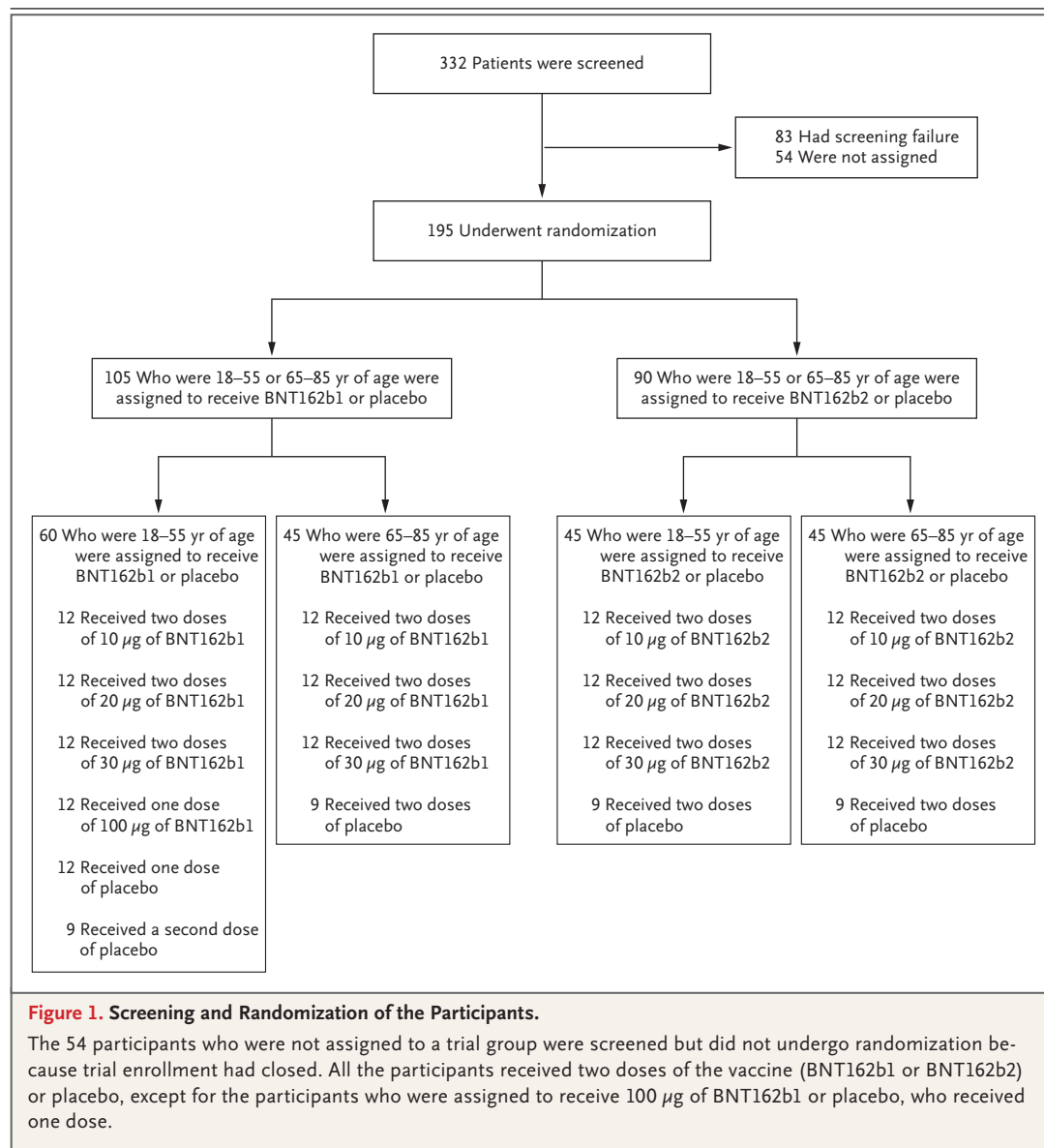


Table 1. Demographic Characteristics of the Participants, According to Vaccine Candidate and Age Group.*

Variable	Participants 18–55 Years of Age						Participants 65–85 Years of Age				
	10 µg	20 µg	30 µg	100 µg	Placebo	Total	10 µg	20 µg	30 µg	Placebo	Total
BNT162b1											
No. of participants	12	12	12	12	12	60	12	12	12	9	45
Sex — no. (%)											
Male	7 (58)	9 (75)	6 (50)	5 (42)	7 (58)	34 (57)	4 (33)	4 (33)	4 (33)	1 (11)	13 (29)
Female	5 (42)	3 (25)	6 (50)	7 (58)	5 (42)	26 (43)	8 (67)	8 (67)	8 (67)	8 (89)	32 (71)
Race — no. (%)†											
White	8 (67)	11 (92)	10 (83)	11 (92)	11 (92)	51 (85)	12 (100)	11 (92)	10 (83)	9 (100)	42 (93)
Black	1 (8)	1 (8)	0	0	0	2 (3)	0	1 (8)	0	0	1 (2)
Asian	3 (25)	0	2 (17)	1 (8)	1 (8)	7 (12)	0	0	2 (17)	0	2 (4)
Hispanic ethnic group — no. (%)†	1 (8)	0	1 (8)	0	0	2 (3)	0	0	0	1 (11)	1 (2)
Age — yr‡											
Mean	29.4±6.4	44.8±8.3	35.8±10.0	38.3±9.3	36.3±11.3	36.9±10.2	69.7±5.4	70.6±4.9	69.9±3.6	68.2±3.0	69.7±4.3
Median (range)	26.5 (24–42)	49.0 (30–54)	33.5 (23–52)	38.0 (25–53)	35.0 (19–54)	35.0 (19–54)	68.5 (65–82)	69.0 (65–81)	69.0 (65–77)	68.0 (65–73)	69.0 (65–82)
BNT162b2											
No. of participants	12	12	12	0	9	45	12	12	12	9	45
Sex — no. (%)											
Male	5 (42)	6 (50)	3 (25)	—	5 (56)	19 (42)	2 (17)	5 (42)	6 (50)	4 (44)	17 (38)
Female	7 (58)	6 (50)	9 (75)	—	4 (44)	26 (58)	10 (83)	7 (58)	6 (50)	5 (56)	28 (62)
Race — no. (%)†											
White	11 (92)	10 (83)	9 (75)	—	9 (100)	39 (87)	12 (100)	12 (100)	12 (100)	9 (100)	45 (100)
Black	0	2 (17)	1 (8)	—	0	3 (7)	0	0	0	0	0
Asian	1 (8)	0	2 (17)	—	0	3 (7)	0	0	0	0	0
Hispanic ethnic group — no. (%)†	1 (8)	1 (8)	0	—	0	2 (4)	0	0	0	0	0
Age — yr‡											
Mean	36.8±12.2	37.6±10.1	37.3±9.8	—	34.4±13.2	36.7±11.0	68.0±2.9	71.0±5.8	68.5±2.8	70.0±3.8	69.3±4.1
Median (range)	37.0 (21–53)	38.0 (23–53)	36.5 (23–54)	—	30.0 (19–53)	37.0 (19–54)	67.0 (65–73)	68.5 (65–81)	68.0 (65–74)	69.0 (65–77)	68.0 (65–81)

* Plus-minus values are means ±SD. Percentages may not total 100 because of rounding.

† Race and ethnic group were reported by the participant.

‡ The age of the participants was the age at the time of the injection.

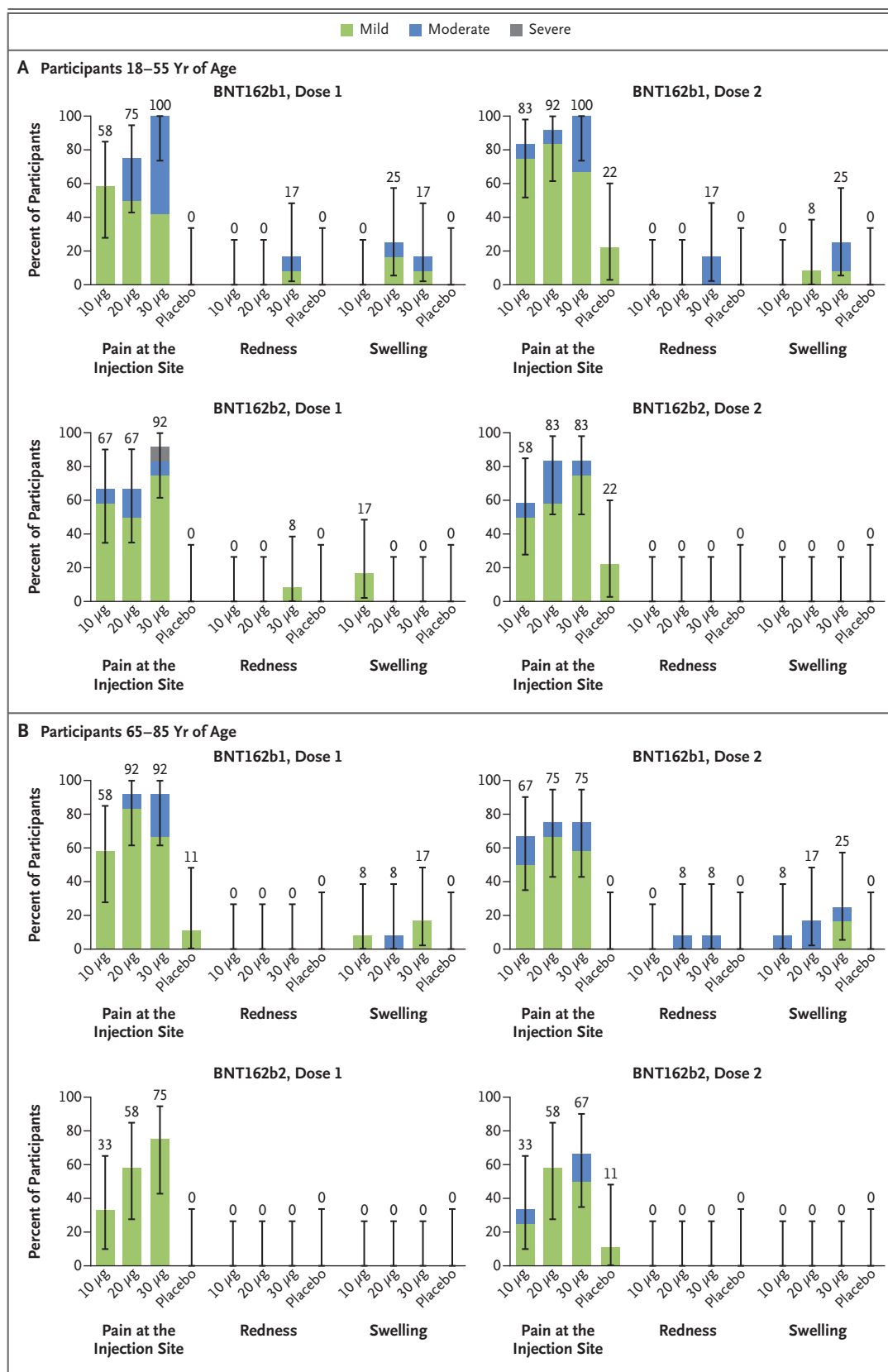


Figure 2 (facing page). Local Reactions Reported within 7 Days after the Administration of Vaccine or Placebo, According to Age Group.

Panel A shows local reactions in participants 18 to 55 years of age, and Panel B those in participants 65 to 85 years of age. Injection-site (local) reactions were recorded in electronic diaries for 7 days after each injection. Pain at the injection site was graded as mild (does not interfere with activity), moderate (interferes with activity), severe (prevents daily activity), or grade 4 (led to an emergency department visit or hospitalization). Redness and swelling were graded as mild (2.0 to 5.0 cm in diameter), moderate (>5.0 to 10.0 cm in diameter), severe (>10.0 cm in diameter), or grade 4 (necrosis or exfoliative dermatitis for redness and necrosis for swelling). I bars represent 95% confidence intervals. The numbers above the I bars show the overall percentage of the participants in each group who reported the specified local reaction. No participant who received either vaccine candidate reported a grade 4 local reaction.

but one, all the participants who underwent randomization received the assigned two doses of vaccine or placebo. Participants 18 to 55 years of age who had been assigned to receive 100 μ g of BNT162b1 or placebo received one dose; the second dose was not administered because of reactogenicity in the participants who received active vaccine.⁵

The majority of participants were White (67 to 100%) and non-Hispanic (89 to 100%) (Table 1). More older women than older men participated. The median age among the younger participants was 35 years in the BNT162b1 group and 37 years in the BNT162b2 group; the median age among the older participants was 69 years and 68 years, respectively.

SAFETY*Local Reactions*

Participants 18 to 55 years of age who received 10 μ g, 20 μ g, or 30 μ g of BNT162b1 reported mild-to-moderate local reactions, primarily pain at the injection site, within 7 days after an injection; the local reactions were more frequent after the second dose.^{2,5} BNT162b1 elicited local reactions in similar proportions of the participants in the younger age group and in the older age group. Among the older participants, mild-to-moderate injection-site pain was reported by 92% after the first dose and by 75% after the second dose (Fig. 2). A similar pattern was observed after vaccination with BNT162b2. No older par-

ticipant who received BNT162b2 reported redness or swelling. No participant who received either BNT162 vaccine candidate reported a grade 4 local reaction.

Systemic Events

Participants 18 to 55 years of age who received 10 μ g, 20 μ g, or 30 μ g of BNT162b1 frequently had mild-to-moderate fever and chills, with 75% of the participants reporting a temperature of 38.0°C or higher after the second 30- μ g dose (Fig. 3; and Fig. S1 in the Supplementary Appendix, available at NEJM.org).⁵ In participants 65 to 85 years of age who received BNT162b1, systemic events were milder than in the younger participants, although many older participants reported fatigue and headache after the first or second dose, and 33% reported a temperature of 38°C or higher after the second dose, including one older participant who reported a fever of 38.9 to 40.0°C (Fig. 3 and Fig. S2). As was observed with local reactions, systemic events were dose-dependent (greater after the second dose than after the first dose) and transient. Symptoms generally peaked by day 2 after vaccination and resolved by day 7.

Systemic events in response to BNT162b2 were milder than those in response to BNT162b1 (Fig. 3 and Figs. S1 and S2). For example, 17% of the participants 18 to 55 years of age and 8% of those 65 to 85 years of age reported fever (≥ 38.0 to 38.9°C) after the second dose of 30 μ g of BNT162b2. Severe systemic events (fatigue, headache, chills, muscle pain, and joint pain) were reported in small numbers of younger recipients of BNT162b2, but no severe systemic events were reported by older recipients of this vaccine candidate. No participant who received either BNT162 vaccine candidate reported a grade 4 systemic event. After the first dose, systemic events that were reported by participants 65 to 85 years of age who received BNT162b2 were similar to those reported by participants who received placebo.

In both age groups and for both vaccine candidates, the use of antipyretic or pain medication increased with increasing dose level and with the number of doses administered. Fewer BNT162b2 recipients than BNT162b1 recipients reported using antipyretic or pain medication.

Adverse Events and Shifts in Laboratory Values

Through 1 month after the receipt of the second dose, adverse events that were considered by the

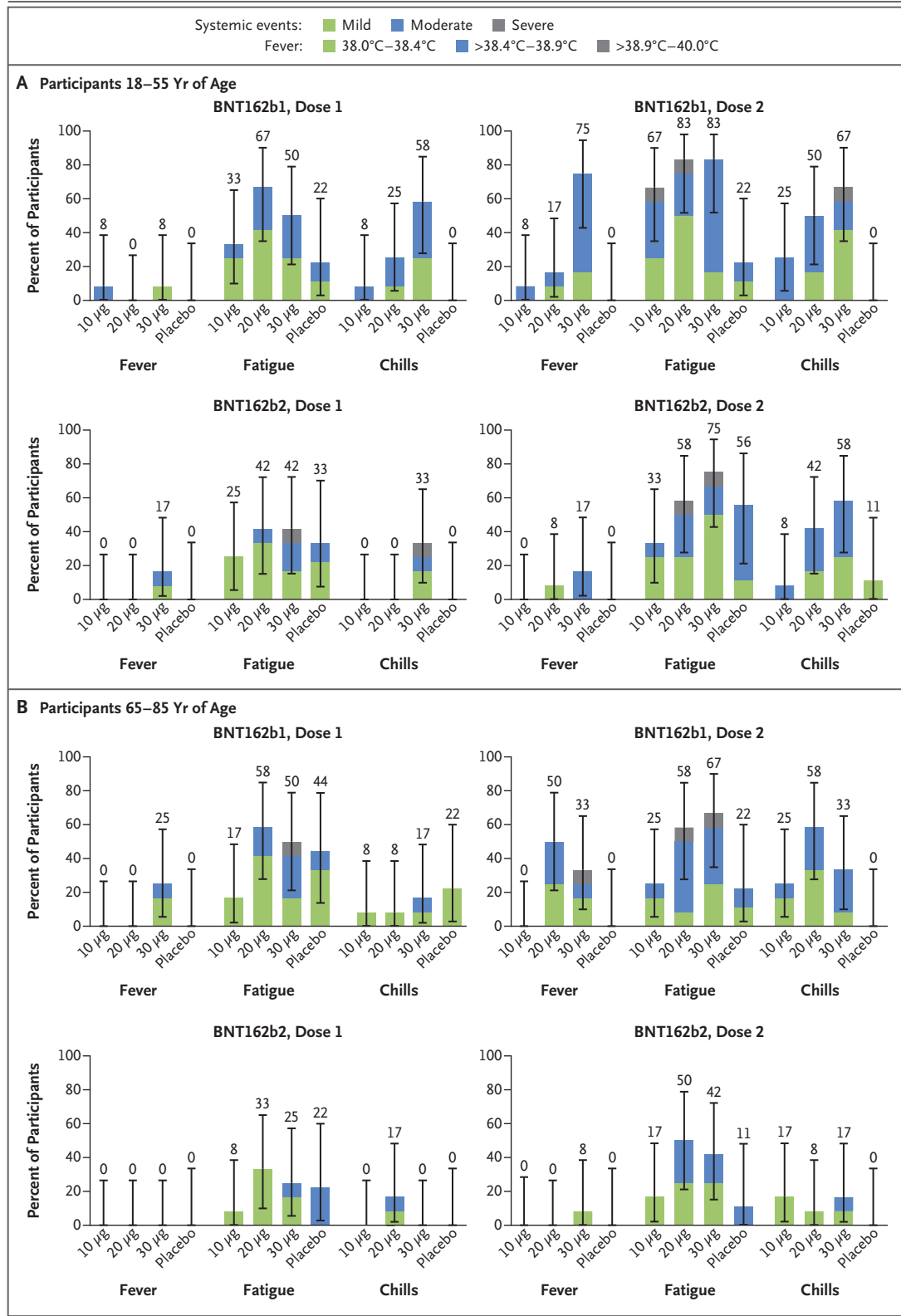


Figure 3 (facing page). Selected Systemic Events Reported within 7 Days after the Administration of Vaccine or Placebo, According to Age Group.

Panel A shows systemic reactions in participants 18 to 55 years of age, and Panel B those in participants 65 to 85 years of age. Data on fever, chills, and fatigue are reported here. (Data on headache, vomiting, diarrhea, muscle pain, and joint pain are reported in Fig. S1.) Data on systemic events were recorded in electronic diaries for 7 days after each injection. The fever scale is shown in the key. Chills and fatigue were graded as being mild (does not interfere with activity), moderate (interferes somewhat with activity), severe (prevents daily activity), or grade 4 (led to an emergency department visit or hospitalization). I bars represent 95% confidence intervals. The numbers above the I bars show the overall percentage of participants in each group who reported the specified systemic event. No participant who received either vaccine candidate reported a grade 4 systemic event or a temperature higher than 40.0°C.

investigators to be related to vaccine or placebo were reported by 50% of the participants 18 to 55 years of age who received 30 μ g of BNT162b1, as compared with 8% of those who received placebo.⁵ Adverse events that were considered to be related to vaccine were reported by 17% of the participants 65 to 85 years of age who received 30 μ g of BNT162b1 and by 25% of the participants 18 to 55 years of age who received 30 μ g of BNT162b2. No participant 65 to 85 years of age who received 30 μ g of BNT162b2 reported a related adverse event (Table S1).

No serious adverse events were reported, and no stopping rules were met as of the time of this report. The largest changes from baseline in laboratory values were transient decreases in lymphocyte counts, which resolved within 1 week after vaccination (Fig. S3) and which were not associated with clinical manifestations.

IMMUNOGENICITY

The serologic responses elicited by BNT162b1 and BNT162b2 were similar (Fig. 4). Two serum samples, both from the group of participants 18 to 55 years of age who received 30 μ g of BNT162b2, were obtained outside the specified time windows (one each at day 28 and day 35) and thus were excluded from the reported immunogenicity analysis. Antigen-binding IgG and virus-neutralizing responses to vaccination with

10 μ g to 30 μ g of BNT162b1 or BNT162b2 were boosted by the second dose in both the younger adults^{2,5} and the older adults. Both vaccines elicited generally lower antigen-binding IgG and virus-neutralizing responses in participants 65 to 85 years of age than in those 18 to 55 years of age. Higher doses appeared to elicit somewhat higher antibody responses.

The highest neutralization titers were measured in samples obtained on day 28 (i.e., 7 days after the second dose) or on day 35 (i.e., 14 days after the second dose). Similar trends were observed for the 50% and 90% neutralizing titers (Fig. S4). The 50% neutralizing GMTs for the two vaccine candidates at the 30- μ g dose level on day 28 or day 35 ranged from 1.7 to 4.6 times the GMT of the convalescent serum panel among participants 18 to 55 years of age and from 1.1 to 2.2 times the GMT of the convalescent serum panel among those 65 to 85 years of age. With 10 to 12 valid results per assay from samples that could be evaluated for each group at each time point, pair-wise comparisons are subject to error and have no clear interpretation.

DISCUSSION

Previously reported data from vaccination with 10 μ g or 30 μ g of BNT162b1 in adults 18 to 55 years of age suggested that it could be a promising Covid-19 vaccine candidate.^{2,5} Consistent with our strategy to evaluate several RNA vaccine candidates and make a data-driven decision to advance the candidate with the best safety and immunogenicity profile, we compared clinical data obtained after vaccination with BNT162b1,^{2,5} which encodes the RBD, with data obtained after vaccination with BNT162b2, which encodes the full-length spike. The data presented here include those that guided our decision to advance BNT162b2 at the 30- μ g dose level to the phase 2–3, international trial to evaluate its safety and efficacy in participants 18 to 85 years of age.

The primary consideration driving this decision was the milder systemic reactogenicity profile of BNT162b2, particularly in older adults, in the context of the similar antibody responses elicited by the two candidate vaccines. Short-lived decreases in postvaccination lymphocyte counts had no associated clinical effect, were observed

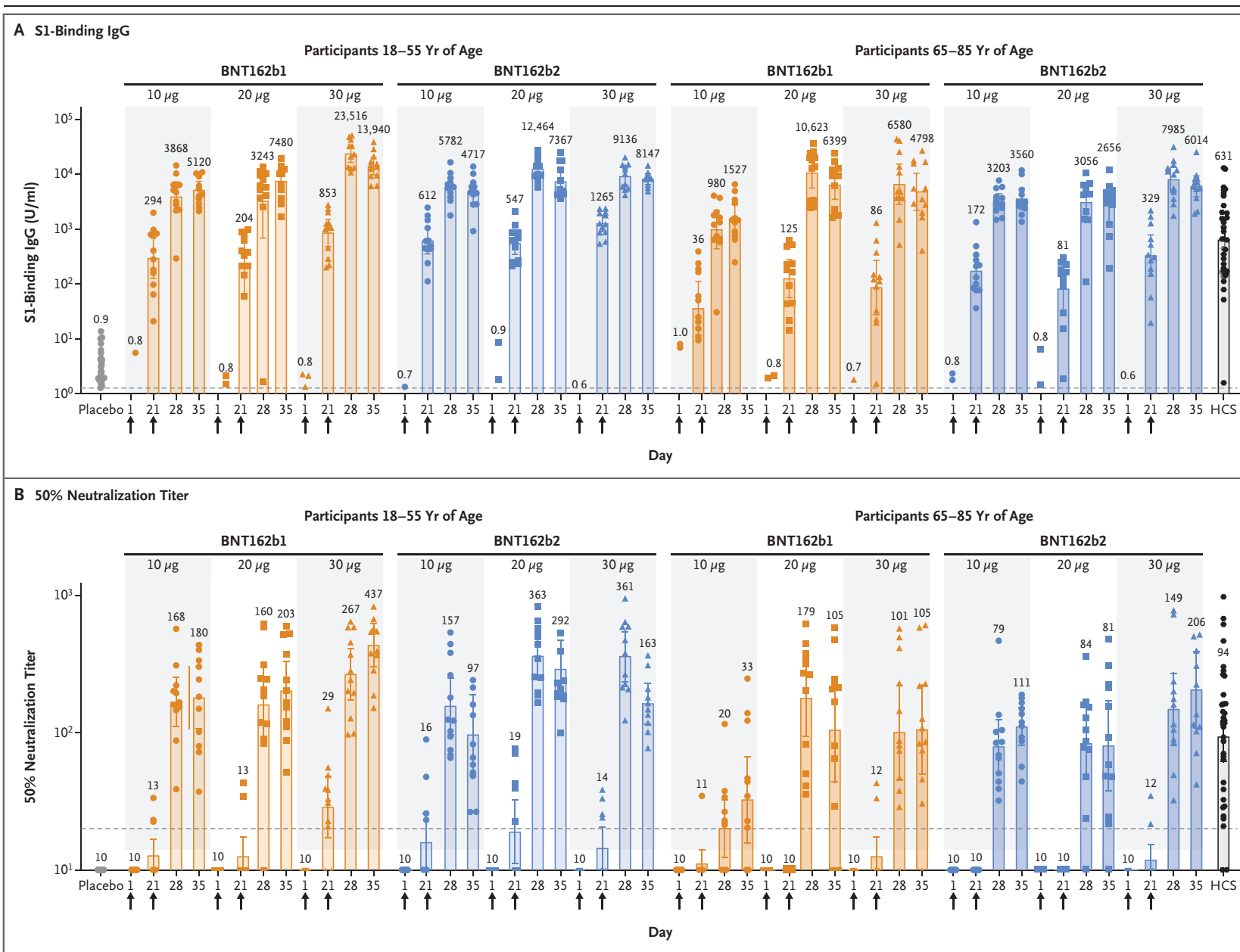


Figure 4 (facing page). Immunogenicity of BNT162b1 and BNT162b2.

Participants in groups of 15 received an injection with the indicated dose levels of one of either of the BNT162 vaccine candidates (12 participants) or placebo (3 participants) on days 1 and 21. Arrows indicate days of vaccination. Responses in the placebo recipients in each of the dose-level groups are combined. Serum samples were obtained before injection (on day 1) and on days 21, 28, and 35 after the first dose. The blood samples obtained on days 28 and 35 are those obtained 7 days and 14 days, respectively, after the second dose. Human coronavirus disease 2019 (Covid-19) or SARS-CoV-2 infection convalescent serum (HCS) samples were obtained from 38 donors at least 14 days after polymerase chain reaction–confirmed diagnosis and at a time when the donors were asymptomatic. Panel A shows the geometric mean concentrations of recombinant S1-binding IgG (lower limit of quantitation, 1.267; dashed line), and Panel B the 50% SARS-CoV-2–neutralizing geometric mean titers (lower limit of quantitation, 20; dashed line). On days that vaccine or placebo was administered, samples were obtained before the injection. Each data point represents a serum sample, and the top of each vertical bar represents the geometric mean with the 95% confidence interval (I bar). Data points associated with placebo, HCS samples, or the 10- μ g dose of vaccine are shown as circles, those for the 20- μ g dose as squares, and those for the 30- μ g dose as triangles. The numbers above the bars show the geometric mean concentration or geometric mean titer in the group. All the vaccine groups had 12 valid results from samples that could be evaluated at each time point except for the following: among participants who received BNT162b2, 11 results from day 28 in younger participants who received 30 μ g, 10 results from day 35 in younger participants who received 30 μ g, and 11 results from day 35 in older participants who received 10 μ g.

across the age groups, and probably reflect a temporary redistribution of lymphocytes from the bloodstream to lymphoid tissues as a functional response to immune stimulation by the vaccine.¹³⁻¹⁶ The immune response and toxicity profile at the selected, relatively low, 30- μ g dose level indicate that the BNT162b2 modRNA vaccine candidate has a favorable balance of reactogenicity and immunogenicity.^{17,18}

The composition of the lipid nanoparticles, the formulation components, or the sequence selection for the vaccine RNA could influence the side-effect profile. The reason for the lower reactogenicity of BNT162b2 than of BNT162b1 is not certain, given that the two vaccine candidates share the same modRNA platform, RNA production and purification processes, and for-

mulation of lipid nanoparticles. They differ in the nucleotide sequences that encode the vaccine antigens and in the overall size of the RNA constructs, which results in a number of RNA molecules in 30 μ g of BNT162b1 that is approximately 5 times as high as that in 30 μ g of BNT162b2. The nucleotide composition of RNA has been reported to affect its immune stimulatory activity and reactogenicity profile, and this is a possible explanation for the differences in these vaccine candidates.¹⁹

The immune responses elicited by BNT162b1 and BNT162b2 were similar. As has been observed with other vaccines and as is probably associated with immunosenescence,^{20,21} the immunogenicity of the two vaccine candidates decreased with age, eliciting lower overall humoral responses in adults 65 to 85 years of age than in those 18 to 55 years of age. Nevertheless, at 7 days and 14 days after the second dose, the 50% and 90% neutralizing GMTs that were elicited by 30 μ g of BNT162b2 in older adults exceeded those of the convalescent serum panel. Antibody responses in both younger and older adults showed a clear benefit of a second dose.

This trial and interim report have several limitations. First, the relative importance of humoral and cellular immunity with regard to protection from Covid-19 has not yet been fully characterized. Although strong cell-mediated immune responses (Th1-biased CD4+ and CD8+) elicited by BNT162b1 have been observed and reported in the German trial,² the cellular immune responses elicited by BNT162b2 are still being studied. Second, although the serum neutralizing responses that were elicited by the vaccine candidates relative to those elicited by natural infection are highly encouraging, the degree of protection against Covid-19 provided by this or any other benchmark is unknown. Third, the phase 1 portion of this trial tested many hypotheses and was not powered to make formal statistical comparisons. Fourth, the human convalescent serum panels that have been used by different vaccine developers are not standardized among laboratories, and each represents a unique distribution of donor characteristics and times of collection. Therefore, the serum panel that we used does not provide a well-controlled benchmark for comparisons of the serologic responses elicited by these two BNT162 vaccine candidates with those elicited by other Covid-19 vaccine candi-

dates. Finally, the participants in this early-stage clinical trial were healthy and had limited racial and ethnic diversity as compared with the general population.

Many of the limitations cited above are being addressed in the international, phase 2–3 portion of this trial. In this later, pivotal part of the trial, we are assessing the safety and efficacy of two doses of 30 μg of BNT162b2 in up to 44,000 participants (randomly assigned in a 1:1 ratio to receive vaccine or placebo) from diverse backgrounds, including persons with stable chronic underlying health conditions, persons at increased risk owing to occupational exposure, and persons from racial and ethnic backgrounds at higher risk for severe Covid-19.²² We are conducting outreach to recruit trial participants from many backgrounds and are using U.S. Census data to locate trial sites in diverse communities.

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Disclosure forms provided by the authors are available with the full text of the article at NEJM.org.

A data sharing statement provided by the authors is available with the full text of this article at NEJM.org.

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